

TED GERSHON

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

B.S. in Information Systems (Minors in Software Engineering, Artificial Intelligence)

Aug 2023 — May 2027

- Relevant Coursework: Data Structures and Algorithms, Computer Systems, ML in Production, Computational Data Science, Principles of Software Construction, Foundations of Software Engineering, Database Design

SKILLS

Languages: Python | SQL | TypeScript/JavaScript | Java | C# | C

Platforms: Databricks | Spark | dbt | PostgreSQL | MongoDB | Redis

Frameworks & Tools: Git | Docker | Next.js | Node.js | React | MLflow | Django | FastAPI

EXPERIENCE

Microsoft

Redmond, WA

Software Engineer Intern

May 2026 — Aug 2026

- Developed a custom Power Query connector to enable unattended refreshes of Excel Online (**153M** MAU) workbooks via Dataflows, executing operations through a live, server-side session via authenticated Graph calls
- Engineered OAuth2 delegated token lifecycle from interactive consent to Fabric-managed (DMTS) credential replay for headless runs, with a TestConnection gate detecting expired tokens before refresh
- Collaborated with connector, dataflow, Excel, DMTS, and OAuth SMEs to design cloud-aware authentication, delegated Graph scopes, and schema metadata modeled after first-party Microsoft connectors

Pennybacker

Austin, TX

Data Engineer Intern

May 2025 — Aug 2025

- Built 3 data models powering client-facing Power BI dashboards for suite managing **\$5.1B AUM** across 4 teams
- Translated real estate and executive dashboard requirements into dbt models, refactoring **14** with Jinja templating to run on Databricks Spark SQL, cutting compile time by up to **85%** and enabling property-level analysis
- Refactored GBDT training pipeline in Python (sklearn, MLflow) to centralize feature engineering across **3 asset classes** and automate logging for **36** configurations, improving consistency and eliminating manual output tracking

PGH Donut Dash

Pittsburgh, PA

Chairman & Lead Developer

Jan 2025 — Present

- Directed development and operations for 501(c)(3) event that has raised **\$265,000+** since 2018 (**12k+** annual site visits), leading site migration to a Next.js/MongoDB monorepo and authoring the full issue backlog
- Integrated Stripe Checkout powering all race registrations and donations, finalizing paid registrations and automating thank-you and confirmation emails from Stripe webhooks via the Gmail REST API (OAuth2)
- Hardened admin console by resolving production bugs and authoring **130+** unit, integration, and end-to-end tests

ERIN

Pittsburgh, PA

Technical Enablement Intern

June 2024 — Aug 2024

- Authored **33** technical guides on SSO authentication, ATS/HRIS integrations, and AI-powered job matching features to support platform adoption for **1.1M+** enterprise users across ERIN's customer portfolio
- Restructured HubSpot knowledge base taxonomy and routing, reducing content navigation from **6-7** clicks to **3**

PROJECTS

BananaGains - FastAPI, Next.js, React, PostgreSQL, Redis, WebSockets

March — April 2026

- Engineered transactional betting engine using PL/pgSQL stored procedures, row-level locking, and ACID-compliant PostgreSQL to prevent race conditions across simultaneous trades, with atomic rollbacks on failure
- Architected real-time market infrastructure with Redis pub/sub fan-out and WebSockets to stream live probability updates, resolution vote counts, and leaderboard changes across clients, with response caching and rate limiting

Odyssey - Python, Django, React, OpenAI, MCP, Mapbox GL JS

February 2026

- Placed **Top 10** out of **278** teams and **#2** in the Creative Arts track at CMU's TartanHacks
- Designed multi-agent MCP orchestration system with a Conductor agent coordinating 4 specialist agents (Archivist, Linguist, Stylist, Librarian) in parallel with real-time performance monitoring
- Engineered PDF-to-map pipeline with AI-powered location extraction that auto-generates interactive GeoJSON markers from uploaded books, rendered on era-themed Mapbox visualizations

LEADERSHIP

ScottyLabs | Head of Logistics

May 2024 — May 2025

- Coordinated Pittsburgh's largest hackathon, TartanHacks, which had **750+** participants from over **18** countries
- Led the first iteration of NOVA, CMU's first GenAI hackathon with **130+** participants and **29** submitted projects